1) There are four pens: Red, Green, Blue and Purple in a desk drawer of which two pens are selected at random one after the other with replacement. State the sample space and the following events.

   a) A : Selecting at least one red pen.

   b) B : Two pens of the same color are not selected.

2) A coin and a die are tossed simultaneously. Enumerate the sample space and the following events.

   a) A : Getting a Tail and an Odd number

   b) B : Getting a prime number

   c) C : Getting a head and a perfect square.

3) Find n(S) for each of the following random experiments.

   a) From an urn containing 5 gold and 3 silver coins, 3 coins are drawn at random

   b) 5 letters are to be placed into 5 envelopes such that no envelop is empty.

   c) 6 books of different subjects arranged on a shelf.

   d) 3 tickets are drawn from a box containing 20 lottery tickets.

4) Two fair dice are thrown. State the sample space and write the favorable outcomes for the following events.

a) A : Sum of numbers on two dice is divisible by 3 or 4.

b) B : Sum of numbers on two dice is 7.

c) C : Odd number on the first die.

d) D : Even number on the first die.

e) Check whether events A and B are mutually exclusive and exhaustive.

f) Check whether events C and D are mutually exclusive and exhaustive.

5) A bag contains four cards marked as 5, 6, 7 and 8. Find the sample space if two cards are drawn at random

a) with replacement

b) without replacement

6) A fair die is thrown two times. Find the probability that

a) sum of the numbers on them is 5

b) sum of the numbers on them is at least 8

c) first throw gives a multiple of 2 and second throw gives a multiple of 3.

d) product of numbers on them is 12.

7) Two cards are drawn from a pack of 52 cards. Find the probability that

a) one is a face card and the other is an ace card

b) one is club and the other is a diamond

c) both are from the same suit.

d) both are red cards

e) one is a heart card and the other is a non heart card

8) Three cards are drawn from a pack of 52 cards. Find the chance that

a) two are queen cards and one is an ace card

b) at least one is a diamond card

c) all are from the same suit

d) they are a king, a queen and a jack

9) From a bag containing 10 red, 4 blue and 6 black balls, a ball is drawn at random. Find the probability of drawing

a) a red ball.

b) a blue or black ball.

c) not a black ball.

10) A box contains 75 tickets numbered 1 to 75. A ticket is drawn at random from the box. Find the probability that,

a) Number on the ticket is divisible by 6

b) Number on the ticket is a perfect square

c) Number on the ticket is prime

d) Number on the ticket is divisible by 3 and 5

11) What is the chance that a leap year, selected at random, will contain 53 sundays?.

12) Find the probability of getting both red balls, when from a bag containing 5 red and 4 black balls, two balls are drawn, i) with replacement ii) without replacement

13) A room has three sockets for lamps. From a collection 10 bulbs of which 6 are defective. At night a person selects 3 bulbs, at random and puts them in sockets. What is the probability that i) room is still dark ii) the room is lit

14) Letters of the word MOTHER are arranged at random. Find the probability that in the arrangement

a) vowels are always together

b) vowels are never together

c) O is at the begining and end with T

d) starting with a vowel and end with a consonant

15) 4 letters are to be posted in 4 post boxes. If any number of letters can be posted in any of the 4 post boxes, what is the probability that each box contains only one letter?

1) First 6 faced die which is numbered 1 through 6 is thrown then a 5 faced die which is numbered 1 through 5 is thrown. What is the probability that sum of the numbers on the upper faces of the dice is divisible by 2 or 3?

2) A card is drawn from a pack of 52 cards. What is the probability that,

   i) card is either red or black?

   ii) card is either black or a face card?

3) A girl is preparing for National Level Entrance exam and State Level Entrance exam for professional courses. The chances of her cracking National Level exam is 0.42 and that of State Level exam is 0.54. The probability that she clears both the exams is 0.11. Find the probability that (i) She cracks at least one of the two exams (ii) She cracks only one of the two (iii) She cracks none

4) A bag contains 75 tickets numbered from 1 to 75. One ticket is drawn at random. Find the probability that,

   a) number on the ticket is a perfect square or divisible by 4

   b) number on the ticket is a prime number or greater than 40

5) The probability that a student will pass in French is 0.64, will pass in Sociology is 0.45 and will pass in both is 0.40. What is the probability that the student will pass in at least one of the two subjects?

6) Two fair dice are thrown. Find the probability that number on the upper face of the first die is 3 or sum of the numbers on their upper faces is 6.

7) For two events A and B of a sample space S, if $P(A) = \frac{3}{8}$, $P(B) = \frac{1}{2}$ and $P(A \cup B) = \frac{5}{8}$.

Find the value of the following.

a) $P(A \cap B)$

b) $P(A' \cap B')$

c) $P(A' \cup B')$

8) For two events A and B of a sample space S, if $P(A \cup B) = \frac{5}{6}$, $P(A \cap B) = \frac{1}{3}$ and $P(B') = \frac{1}{3}$, then find P(A).

9) A bag contains 5 red, 4 blue and an unknown number m of green balls. If the probability of getting both the balls green, when two balls are selected at random is $\frac{1}{7}$, find m.

10) Form a group of 4 men, 4 women and 3 children, 4 persons are selected at random. Find the probability that, i) no child is selected ii) exactly 2 men are selected.

11) A number is drawn at random from the numbers 1 to 50. Find the probability that it is divisible by 2 or 3 or 10.

1) A bag contains 3 red marbles and 4 blue marbles. Two marbles are drawn at random without replacement. If the first marble drawn is red, what is the probability the second marble is blue?

2) A box contains 5 green pencils and 7 yellow pencils. Two pencils are chosen at random from the box without replacement. What is the probability that both are yellow?

3) In a sample of 40 vehicles, 18 are red, 6 are trucks, of which 2 are red. Suppose that a randomly selected vehicle is red. What is the probability it is a truck?

4) From a pack of well-shuffled cards, two cards are drawn at random. Find the probability that both the cards are diamonds when

   i) first card drawn is kept aside

   ii) the first card drawn is replaced in the pack.

5) A, B, and C try to hit a target simultaneously but independently. Their respective probabilities of hitting the target are $\frac{3}{4}$, $\frac{1}{2}$ and $\frac{5}{8}$. Find the probability that the target

   a) is hit exactly by one of them

   b) is not hit by any one of them

   c) is hit

   d) is exactly hit by two of them

6) The probability that a student X solves a problem in dynamics is $\frac{2}{5}$ and the probability that student Y solves the same problem is $\frac{1}{4}$. What is the probability that

i) the problem is not solved

ii) the problem is solved

iii) the problem is solved exactly by one of them

7) A speaks truth in 80% of the cases and B speaks truth in 60% of the cases. Find the probability that they contradict each other in narrating an incident.

8) Two hundred patients who had either Eye surgery or Throat surgery were asked whether they were satisfied or unsatisfied regarding the result of their surgery.

The follwoing table summarizes their response.

Surgery	Satisfied	Unsatisfied	Total
Throat	70	25	95
Eye	90	15	105
Total	160	40	200

If one person from the 200 patients is selected at random, determine the probability

a) that the person was satisfied given that the person had Throat surgery

b) that person was unsatisfied given that the person had eye surgery

c) the person had Throat surgery given that the person was unsatisfied

9) Two dice are thrown together. Let A be the event 'getting 6 on the first die' and B be the event 'getting 2 on the second die'. Are the events A and B independent?

10) The probability that a man who is 45 years old will be alive till he becomes 70 is $\frac{5}{12}$. The probability that his wife who is 40 years old will be alive till she becomes 65 is $\frac{3}{8}$. What is the probability that, 25 years hence,

a) the couple will be alive

b) exactly one of them will be alive

c) none of them will be alive

d) at least one of them will be alive

11) A box contains 10 red balls and 15 green balls. Two balls are drawn in succession without replacement. What is the probability that,

a) the first is red and the second is green?

b) one is red and the other is green?

12) A bag contains 3 yellow and 5 brown balls. Another bag contains 4 yellow and 6 brown balls. If one ball is drawn from each bag, what is the probability that,

a) both the balls are of the same color?

b) the balls are of different color?

13) An urn contains 4 black, 5 white and 6 red balls. Two balls are drawn one after the other without replacement. What is the probability that at least one of them is black?

14) Three fair coins are tossed. What is the probability of getting three heads given that at least two coins show heads?

15) Two cards are drawn one after the other from a pack of 52 cards without replacement. What is the probability that both the cards drawn are face cards?

16) Bag A contains 3 red and 2 white balls and bag B contains 2 red and 5 white balls. A bag is selected at random, a ball is drawn and put into the other bag, and then a ball is drawn from that bag. Find the probability that both the balls drawn are of same color.

17) (Activity) : A bag contains 3 red and 5 white balls. Two balls are drawn at random one after the other without replacement. Find the probability that both the balls are white.